Matthew Butrovich

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Summary

Database management systems (DBMS) expert with a deep understanding of high-performance distributed data processing. Strong background in transactional and analytical DBMS design (query execution, storage management, concurrency control, data formats), data lakes (Iceberg, Parquet), Linux internals (eBPF, networking), storage systems (distributed, devices, file systems), performance engineering (benchmarking, observability), systems for machine learning, and machine learning for systems. Committed to lifelong learning, peer mentorship, and contributing to scientific and open source communities.

Professional Experience

2024 – **Apple** — Washington, DC
Senior Distributed Database Engineer
- Enhancing Apple's data platform by contributing to Apache DataFusion Comet, Apache Spark, Apache Iceberg, Apache DataFusion, Apache Parquet, and Apache Arrow.

2009 – 2017 **Western Digital Corporation** — Irvine, CA
Lead Technician V
- Led a team of 17 technicians verifying firmware for HDD and SSD storage devices.
- Promoted four times from initial role as Technician II while achieving B.Sc. degree.

Education

2019 – 2024 **Carnegie Mellon University** — Pittsburgh, PA
Doctor of Philosophy (Ph.D.), Computer Science
- Advisor: Andy Pavlo
- Thesis: "On Embedding Database Management System Logic in Operating Systems via Restricted Programming Environments"

2017 – 2018 **Carnegie Mellon University** — Pittsburgh, PA
Master of Science (M.Sc.), Computer Science

2013 – 2017 **University of California, Irvine** — Irvine, CA
Bachelor of Science (B.Sc.), *cum laude*, Computer Science and Engineering

2010 – 2013 **Irvine Valley College** — Irvine, CA
Certificate of Achievement, Computer Science: Computer Languages

Selected Publications

2025 **Matthew Butrovich**, Samuel Arch, Wan Shen Lim, William Zhang, Jignesh M. Patel, and Andrew Pavlo. "BPF-DB: A Kernel-Embedded Transactional Database Management System For eBPF Applications". *SIGMOD Conference.*

2023 **Matthew Butrovich**, Karthik Ramanathan, John Rollinson, Wan Shen Lim, William Zhang, Justine Sherry, and Andrew Pavlo. "Tigger: A Database Proxy That Bounces With User-Bypass". *Proc. VLDB Endow.*

2022 **Matthew Butrovich**, Wan Shen Lim, Lin Ma, John Rollinson, William Zhang, Yu Xia, and Andrew Pavlo. "Tastes Great! Less Filling! High Performance and Accurate Training Data Collection for Self-Driving Database Management Systems". *SIGMOD Conference.*

Skills

Programming Languages: C, C++, eBPF, Java, Python, Rust, Scala
Development Tools: bpftrace, Claude, GDB/LLDB, Git, Intel VTune, perf

Selected Talks

2025 **CS 224P: Big Data Systems**
Lecture on building petabyte-scale analytics systems

2023 **15-445/645: Database Systems**
Lecture on join algorithms

2019 **15-445/645: Database Systems**
Lecture on two-phase locking concurrency control

Service

To the Open Source Community

2025 – Committer, Apache DataFusion, *The Apache Software Foundation.*

To the Profession

2026 Member, *ACM SIGMOD 2027 Program Committee.*

2026 Member, *VLDB 2027 Program Committee.*

2025 Member, *ACM SIGMOD 2026 Industry Track Program Committee.*

2025 Member, *ACM SIGMOD 2026 Program Committee.*

2025 Reviewer, *ACM Transactions on Database Systems (TODS).*

2024 Member, *eBPF Summit 2024 Program Committee.*

2023 Member, *eBPF Summit 2023 Program Committee.*

2023 Member, *ACM SIGMOD 2023 Artifacts & Reproducibility Committee.*

Projects

2024 **BPF-DB**
ACID-compliant embedded DBMS for kernel-space eBPF applications.
Role: Sole developer. Wrote eBPF programs in C for storage management (variable length key-value store), transaction management (serializable isolation level), and write-ahead logging. Implemented two sample applications built on BPF-DB: (1) Redis-protocol transactional key-value store that matches state-of-the-art approaches (Dragonfly), and (2) Voter workload comparing serializable transaction performance against VoltDB, achieving 43% higher throughput.

2023 **Tigger**
PostgreSQL-compatible DBMS proxy based on PgBouncer that bypasses user-space via eBPF.
Role: Sole developer. Wrote eBPF programs in C that attach at the socket and Traffic Control layers of the Linux networking stack to apply PostgreSQL protocol logic. This design reduced CPU utilization by up to 42%, and lowered transaction latencies by up to 29%.

2018 – 2022 **NoisePage**
PostgreSQL-compatible in-memory DBMS designed for autonomous operation.
Role: Founding developer responsible for HyPer-style MVCC over Apache Arrow-compatible memory format, PostgreSQL protocol (Simple and Extended Query), and training data collection. Wrote a C++ version of TPC-C to exercise the storage layer for performance and correctness, ported the Bw-Tree from Peloton, and contributed to system catalog design.